TEST PLAN DOCUMENT

Web-Based ML Model Evaluator

1. Purpose and Scope

This test plan covers the functional requirements defined in the SRS for the Web-Based ML Model Evaluator. Test cases are grouped by functional module and reference the SRS requirement IDs. Actual Result and Test Result are left blank until execution.

2. Test Case ID Convention

Test Case ID format: <Requirement-ID>-<Test-Type>-<Sequence>, for example FR-2.1-UT-1.

- UT = Unit Testing; IT = Integration Testing; ST = System Testing; NFT = Non-Functional Testing.
- The requirement identifier must match the SRS or Security Requirements Document.
- Actual Result and Test Result are recorded only after the test is executed.

3. Testing Summary

#	Module	Functional Requirements	Test Cases
1	3.1 User Access and Session Management	FR-1.1, FR-1.2, FR-1.3, FR-1.4	7
2	3.2 Dataset Management	FR-2.1, FR-2.2, FR-2.3, FR-2.4, FR-2.5	6
3	3.3 Data Preparation	FR-3.1, FR-3.2, FR-3.3	6
4	3.4 Model Configuration and Training	FR-4.1, FR-4.2, FR-4.3	6
5	3.5 Evaluation, Comparison, and Export	FR-5.1, FR-5.2, FR-5.3, FR-5.4	7
6	3.6 Run History and Service Interface	FR-6.1, FR-6.2, FR-6.3	5
7	3.7 Administration and Instructor Access	FR-7.1, FR-7.2, FR-7.3, FR-7.4	8

3.1 User Access and Session Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.1-UT-1	3.1 User Access and Session Management	Register with valid required fields	Registration page available.	Enter valid required fields and submit.	Valid unique test account	Account is created.		
FR-1.1-UT-2	3.1 User Access and Session Management	Reject registration with missing required field	Registration page available.	Leave a required field blank and submit.	Missing required field	Validation error appears; account is not created.		
FR-1.2-UT-1	3.1 User Access and Session Management	Login with valid credentials	Registered account exists.	Enter valid credentials and submit.	Valid credentials	Authenticated session is created.		
FR-1.2-UT-2	3.1 User Access and Session Management	Reject invalid credentials	Login page available.	Enter incorrect password and submit.	Incorrect password	Login is denied; no authenticated session is created.		
FR-1.3-ST-1	3.1 User Access and Session Management	Logout invalidates active session	User logged in.	Log out, then reuse the prior session on a protected route.	Prior session	Protected access is denied.		
FR-1.4-ST-1	3.1 User Access and Session Management	Deny anonymous protected-resource access	User logged out.	Request protected dataset/run endpoint.	No session	Access is denied; no protected data returned.		
FR-1.4-ST-2	3.1 User Access and Session Management	Prevent cross-user resource access	Two users own separate datasets/runs.	Sign in as User A and request User B's resource ID.	User B resource ID	Request is denied.		

3.2 Dataset Management

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.1-IT-1	3.2 Dataset Management	Upload valid CSV	User logged in.	Select and upload a valid CSV.	CSV with header and rows	Dataset is accepted and associated with the user.		
FR-2.2-UT-1	3.2 Dataset Management	Reject unsupported file type	Upload page available.	Upload a non-CSV file.	PDF/image file	File is rejected with validation feedback.		

FR-2.2-UT-2	3.2 Dataset Management	Reject empty or malformed CSV	Upload page available.	Upload empty and malformed CSV fixtures.	Empty/malformed CSV	Invalid input is rejected.		
FR-2.3-ST-1	3.2 Dataset Management	Preview column names and sample rows	Valid dataset uploaded.	Open dataset preview.	Known CSV	Headers and bounded row sample match input.		
FR-2.4-ST-1	3.2 Dataset Management	Verify dataset statistics	Valid CSV with known rows, columns, types, and missing values.	Open statistics view.	Small known CSV	Displayed available statistics match the data.		
FR-2.5-IT-1	3.2 Dataset Management	Replace dataset without cross-user exposure	User has an existing dataset.	Upload a second dataset and inspect metadata/ownership.	Two valid CSVs	New/replacement dataset is associated with the current user.		

3.3 Data Preparation

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-3.1-UT-1	3.3 Data Preparation	Accept supported preprocessing configuration	Valid dataset selected.	Choose a supported option and submit.	Supported option	Configuration is accepted.		
FR-3.1-UT-2	3.3 Data Preparation	Reject invalid preprocessing configuration	Dataset selected.	Submit unsupported option.	Invalid option	Configuration is rejected with feedback.		
FR-3.2-UT-1	3.3 Data Preparation	Select valid target column	Dataset contains target column.	Choose an existing usable column.	Existing column	Target selection is saved.		
FR-3.2-UT-2	3.3 Data Preparation	Reject nonexistent target column	Dataset selected.	Submit a column name not in the dataset.	Nonexistent column	Configuration is rejected.		
FR-3.3-UT-1	3.3 Data Preparation	Accept supported train-test split	Dataset and target selected.	Choose supported split and submit.	Supported proportion	Split is accepted and partitions are usable.		
FR-3.3-UT-2	3.3 Data Preparation	Reject invalid train-test split	Dataset and target selected.	Enter unsupported proportion.	Invalid proportion	Split is rejected.		

3.4 Model Configuration and Training

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.1-UT-1	3.4 Model Configuration and Training	Select compatible supported algorithm	Dataset/task configuration available.	Choose a supported compatible algorithm.	Supported algorithm	Selection is saved.		
FR-4.1-UT-2	3.4 Model Configuration and Training	Reject unsupported algorithm	Dataset/task configuration available.	Submit unsupported algorithm identifier to backend.	Unsupported identifier	Backend rejects selection.		
FR-4.2-IT-1	3.4 Model Configuration and Training	Train model using configured experiment	Valid dataset, target, split, preprocessing, algorithm.	Start training and wait for completion.	Small valid dataset	Training uses the selected configuration and produces a run.		
FR-4.2-IT-2	3.4 Model Configuration and Training	Prevent training with invalid configuration	Required configuration invalid.	Start training.	Invalid target/split	Training does not start.		
FR-4.3-ST-1	3.4 Model Configuration and Training	Record successful run configuration	Valid experiment configured.	Train and open run details.	Valid experiment	Run status/configuration/results are associated correctly.		
FR-4.3-ST-2	3.4 Model Configuration and Training	Preserve earlier run after failure	A completed run exists.	Trigger controlled failure and reopen prior run.	Invalid experiment	Failure is reported; earlier run remains unchanged.		

3.5 Evaluation, Comparison, and Export

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-5.1-UT-1	3.5 Evaluation, Comparison, and Export	Display applicable classification metrics	Completed classification run exists.	Open run results.	Classification run	Applicable classification metrics are shown.		
FR-5.1-UT-2	3.5 Evaluation, Comparison, and Export	Display applicable regression metrics	Completed regression run exists.	Open run results.	Regression run	Applicable regression metrics are shown.		
FR-5.2-ST-1	3.5 Evaluation, Comparison, and Export	Display supported evaluation visualization	Run supports a visualization.	Open evaluation results.	Compatible run	Supported visualization is displayed.		

FR-5.3-ST-1	3.5 Evaluation, Comparison, and Export	Compare compatible model runs	Two accessible comparable runs exist.	Select both and compare.	Two compatible runs	Comparable metrics/configuration are displayed.		
FR-5.3-ST-2	3.5 Evaluation, Comparison, and Export	Handle incompatible run comparison	Runs have incompatible task types.	Select both and compare.	Classification and regression runs	System prevents misleading comparison and explains limitation.		
FR-5.4-IT-1	3.5 Evaluation, Comparison, and Export	Download supported artifact/report	Completed run has artifact/report.	Select run and download supported output.	Supported format	Correct authorized output is returned.		
FR-5.4-IT-2	3.5 Evaluation, Comparison, and Export	Handle missing artifact or unsupported format	Artifact missing or format unsupported.	Request download.	Unsupported format/missing artifact	Clear error is shown.		

3.6 Run History and Service Interface

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-6.1-ST-1	3.6 Run History and Service Interface	View previous run history	User has prior runs.	Open run history.	Known prior runs	Available statuses/configurations/metrics are shown accurately.		
FR-6.1-ST-2	3.6 Run History and Service Interface	Isolate run history by user	Two users have separate runs.	Open history as User A.	Separate user runs	Only authorized User A runs are visible.		
FR-6.2-IT-1	3.6 Run History and Service Interface	Verify valid API response structure	Backend endpoint available.	Submit valid request and inspect response.	Valid request	Documented success status and response structure are returned.		
FR-6.2-IT-2	3.6 Run History and Service Interface	Verify invalid API response structure	Backend endpoint available.	Submit invalid request and inspect response.	Invalid payload	Controlled structured error is returned.		
FR-6.3-ST-1	3.6 Run History and Service Interface	Show actionable operation failure	Controlled failure can be triggered.	Trigger invalid upload/configuration or failed export.	Malformed CSV/invalid config	Understandable error is shown without sensitive internal details.		

3.7 Administration and Instructor Access

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-7.1-ST-1	3.7 Administration and Instructor Access	Admin views supported user-management functions	Admin account and test users exist.	Sign in as Admin and open user management.	Test accounts	Supported user records/actions are displayed.		
FR-7.1-ST-2	3.7 Administration and Instructor Access	Deny regular user user-management access	Regular user exists.	Call admin route/API as regular user.	Regular-user session	Access is denied.		
FR-7.2-ST-1	3.7 Administration and Instructor Access	Admin views dataset metadata	Admin and stored datasets exist.	Open dataset management.	Known test datasets	Metadata is displayed accurately.		
FR-7.2-ST-2	3.7 Administration and Instructor Access	Preserve ownership during dataset administration	Multiple users' datasets and runs exist.	Perform supported test action and inspect associations.	Two users' resources	Ownership/run references remain consistent.		
FR-7.3-ST-1	3.7 Administration and Instructor Access	Admin views activity logs read-only	Admin and activity events exist.	Perform upload/training and open logs.	Known test events	Available events are shown read-only.		
FR-7.3-ST-2	3.7 Administration and Instructor Access	Deny non-Admin log access	Regular user and Instructor accounts exist.	Request logs UI/API directly as non-Admin.	Non-Admin session	Access is denied.		
FR-7.4-ST-1	3.7 Administration and Instructor Access	Instructor views authorized results read-only	Instructor and authorized result exist.	Sign in as Instructor and open result.	Authorized run	Result is displayed without write access.		
FR-7.4-ST-2	3.7 Administration and Instructor Access	Deny Instructor changes to datasets/training/results	Instructor account exists.	Attempt upload, training, or result modification.	Instructor session	Unauthorized write operation is denied.		